

Anthropometric Measurements for Upper Limb Prosthetic Modelling: A Comprehensive Review

Executive Summary

The design of functional, comfortable, and aesthetically acceptable upper limb prostheses is fundamentally reliant on accurate anthropometric data. This review synthesizes current research on the key measurements of the hand, fingers, thumb, wrist, and forearm that are most critical for prosthetic modeling and personalization. Key dimensions such as hand length, hand breadth, phalanx lengths, and various circumferences form the basis of prosthetic design, influencing everything from kinematic accuracy to socket fit. While traditional measurement techniques like calipers remain prevalent, modern methods including 3D scanning, CT, and MRI offer enhanced precision, speed, and the ability to capture complex surface geometry, which is crucial for creating patient-specific devices. Despite these advancements, significant gaps persist, particularly the lack of comprehensive anthropometric databases for pediatric populations and amputees, and a need for standardized measurement protocols to improve design consistency and comparability. These findings underscore the importance of integrating detailed, population-specific anthropometric data into parametric CAD workflows to advance the development of customized and effective upper limb prostheses.

Measurement Taxonomy by Body Region

Accurate prosthetic design begins with a comprehensive understanding of the dimensions of the human upper limb. These measurements are categorized by body region to ensure all aspects of form and function are considered during the modeling process.

HAND

The overall dimensions of the hand provide the foundational parameters for sizing the entire prosthesis, ensuring it is proportional to the user's body.

- **Hand Length:** Defined as the distance from the wrist crease (stylium landmark) to the tip of the middle finger, this is a primary parameter for scaling the overall prosthesis. Studies report mean hand lengths of 18.82 cm for males and 17.46 cm for females in a Northwest Mexican population, and 19.17 cm for males and 17.33 cm for females in an Egyptian cohort. This measurement is a fundamental input for CAD models and is often included in standardized measurement lists like ISO 7250.
- **Hand Breadth:** Measured across the metacarpal joints (metacarpal II to metacarpal V), this dimension is critical for determining the width of the prosthetic palm. Research has identified it as one of the most important parameters for parametric hand design. Reported mean values include 8.79 cm for males and 7.80 cm for females.
- **Palm Length:** This is the distance from the base of the middle finger to the wrist crease and is essential for sizing the palm component of a prosthesis.
- **Hand Circumference/Girth:** Measured around the metacarpals, this parameter is important for socket design and ensuring a secure fit.
- **Hand Thickness:** The measurement through the hand at the metacarpals is used to ensure the prosthetic has a realistic and functional volume.

FINGERS

Individual finger and phalanx dimensions are crucial for creating kinematically accurate mechanisms that can replicate natural grip patterns.

- **Individual Finger Lengths:** The lengths of the index, middle, ring, and little fingers are measured from the metacarpophalangeal (MCP) joint to the fingertip. One study reported average lengths of 63.9 mm (index), 70.7 mm (middle), 65.5 mm (ring), and 53.3 mm (little finger). These are fundamental inputs for parametric CAD models to customize finger size.
- **Phalanx Lengths (Proximal, Middle, Distal):** The lengths of the individual bone segments of each finger are critical for designing linkage mechanisms and ensuring the joints bend at anatomically correct locations. Studies have used radiography and dissection to measure these bone lengths directly. For example, one prosthetic design used phalanx lengths of 50 mm (proximal), 35 mm (middle), and 30 mm (distal), while another used a ratio of 17.3:29.2:44.5 for distal, intermediate, and proximal phalanges, respectively.
- **Finger Breadth and Depth:** Measured at the proximal and distal interphalangeal joints, these dimensions are used to model the finger's volume and shape, which impacts the aesthetic appearance and the ability to grasp small objects. One study reported a mean breadth of the proximal phalanx of the middle finger as 0.94 cm.
- **Interphalangeal Joint Spacing:** The distances between the DIP, PIP, and MCP joints are derived from phalanx length data and are essential for placing the mechanical joints of the prosthesis correctly.

THUMB

The thumb's unique geometry and opposition are vital for dexterity, making its accurate measurement a priority in prosthetic design.

- **Thumb Length:** Measured from the thumb tip to its base, this is a key parameter for ensuring effective opposition and grasping. A study of young adults reported an average thumb length of 49.5 mm.
- **Thumb Phalanx and Metacarpal Lengths:** The lengths of the thumb's bone segments are used to model its kinematic chain.
- **Thumb Breadth:** This measurement informs the overall size and shape of the thumb mechanism.
- **Opposition Angle and Carpometacarpal (CMC) Joint Geometry:** While not a simple linear measurement, the geometry of the CMC joint and the resulting opposition angle are critical for replicating the thumb's unique multi-degree-of-freedom movement, which is essential for various grip types.

WRIST

The wrist serves as the interface between the hand and forearm, and its dimensions are critical for both socket connection and joint mechanism design.

- **Wrist Breadth, Circumference, and Depth:** These measurements are fundamental for designing the socket or wrist connector of a transradial prosthesis. Mean wrist girths of 14.83 cm for females and 15.57 cm for males have been reported. These measurements are defined in standards like ISO 7250-1.

FOREARM

For transradial amputees, the geometry of the residual limb is the most critical factor for a successful prosthetic fitting.

- **Forearm Length:** The length from the elbow to the wrist or the length of the residual limb is used to scale the prosthetic forearm component.

- **Forearm Circumference:** Measurements at multiple points along the forearm (proximal, mid, distal) are essential for designing a custom-fitted socket that distributes pressure evenly and provides stable suspension.
- **Cross-sectional Shape:** Modern workflows use 3D scanning or CT/MRI to capture the non-circular cross-sectional shape of the residual limb, which is critical for creating a comfortable and functional socket.

Measurement Methods

A variety of methods are employed to capture anthropometric data, ranging from traditional manual techniques to advanced digital imaging technologies. The choice of method often involves a trade-off between accuracy, cost, time, and the complexity of the data required.

The table below summarizes and compares the most common measurement methods.

Measurement Method	Description	Advantages	Limitations & Accuracy
Traditional (Manual)	Use of tools like calipers, measuring tapes, and goniometers to take direct physical measurements.	Low cost, portable, widely understood and accessible.	Prone to operator error, time-consuming for comprehensive measurements, captures limited 2D data, difficult to measure complex shapes.
3D Scanning	Non-contact optical methods (structured light or laser) capture surface geometry as a point cloud, which is converted to a 3D model.	Fast data acquisition (often minutes), high accuracy, captures complex 3D surface geometry, cost-effective.	Can struggle with reflective or transparent surfaces. Accuracy can vary (0.05 mm to 4 mm), with some studies reporting volumetric deviation of 2-9%.
CT/MRI	Medical imaging techniques that provide detailed internal and external anatomical data, including bone and soft tissue structures.	Extremely high accuracy and detail, captures internal bone geometry ideal for kinematic modeling, considered a gold standard for volumetric accuracy.	High cost, slow acquisition time (~30 minutes), radiation exposure (CT), and limited accessibility. MRI has shown the lowest deviation in comparative studies.
Radiography	X-ray imaging used to measure bone lengths and joint geometry from digitized images.	Provides accurate bone dimensions, useful for determining phalanx and metacarpal lengths.	Provides 2D projections, radiation exposure, does not capture soft tissue or surface geometry.
Photogrammetry	Uses multiple photographs from different angles to reconstruct a 3D model or take 2D measurements.	Very low cost (uses standard cameras), portable, and can be used for remote measurement.	Less accurate than dedicated 3D scanners, requires careful calibration and lighting, computationally intensive post-processing.

Measurement Method	Description	Advantages	Limitations & Accuracy
Image Processing	Software-based methods using algorithms (e.g., OpenCV, Mediapipe) to detect landmarks and measure dimensions from 2D or 3D images.	Can automate measurements, enables remote data capture via webcam or smartphone, low cost.	Accuracy is dependent on image quality, lighting, and algorithm robustness; often less precise than direct or 3D scanning methods.

Population Variability and Databases

Anthropometric dimensions of the upper limb exhibit significant variability across different populations, influenced by factors such as sex, age, and geographic origin. Designing effective prostheses requires an understanding of this variability to ensure devices can accommodate a wide range of users. For instance, studies consistently show that male hand dimensions are larger on average than female dimensions. Age is another critical factor, with different data required for pediatric users, whose anatomy changes rapidly, compared to adults.

To address this variability, designers often rely on large-scale anthropometric databases. Key resources include:

- **ANSUR (Anthropometric Survey of U.S. Army Personnel):** A major database providing comprehensive body measurements for military populations, which has historically been a common reference in design.
- **CAESAR (Civilian American and European Surface Anthropometry Resource):** This database contains 3D body scans of thousands of individuals from North America and Europe, offering rich data for modern digital design workflows.
- **ISO 7250:** This international standard provides definitions for body measurements and statistical data from various national populations. ISO/TR 7250-2 provides statistical summaries from different countries, while ISO 7250-1 defines the measurement protocols.
- **National Surveys:** Various countries conduct their own anthropometric surveys to gather data specific to their populations, which can be more accurate for local design applications.

When designing for a broad user base, a common practice is to accommodate the range from the 5th percentile female to the 95th percentile male to cover approximately 90% of the target population.

Application to Prosthetic Design Workflows

Anthropometric measurements are not merely static data points; they are integral inputs that drive modern, personalized prosthetic design workflows from conception to fabrication.

- **Parametric CAD:** This is a primary application where key anthropometric values (e.g., hand length, finger width, palm breadth) are used as controlling parameters in a CAD model. By linking the model's geometry to these parameters, designers can rapidly generate a customized prosthesis for an individual by simply inputting their measurements. This approach is far superior to uniform scaling, which often results in poor fit due to the weak correlation between different body dimensions like finger length and width.
- **Socket Design:** For transradial and transhumeral prostheses, the design of a comfortable and stable socket is paramount. 3D scans or CT/MRI data of the residual limb are used to create a precise digital model of the stump. This model serves as the basis for designing a custom socket that conforms perfectly to the user's anatomy, ensuring even pressure distribution and preventing discomfort or injury.

- **Linkage Mechanism Sizing:** The lengths of phalanges and the distances between joints directly inform the design of kinematic mechanisms, such as four-bar linkages, used to replicate finger movement. Accurate measurements ensure that the prosthetic finger's range of motion closely mimics the natural path, which is crucial for effective grasping.
- **3D Printing Customization:** Additive manufacturing enables the direct fabrication of these parametrically designed, personalized components.
- **Finite Element Analysis (FEA):** Anthropometric data is used to create anatomically correct models for FEA simulations. These simulations can predict stress distributions, deformations under load, and pressure at the socket-limb interface, allowing designers to optimize the structural integrity and comfort of the prosthesis before fabrication.
- **Kinematic Modeling:** Measurements of bone lengths and joint locations extracted from sources like radiographs or CT scans are used to create and validate kinematic models of the hand. These models are essential for designing control systems and ensuring that the prosthetic hand moves in a natural and intuitive way.

Statistical Approaches

To manage the complexity of the numerous variables involved in hand anthropometry, researchers employ statistical techniques to identify the most critical dimensions and predict others.

- **Factor Analysis and Principal Component Analysis (PCA):** These methods are used for dimensionality reduction. By analyzing a large set of correlated measurements, PCA can identify a smaller set of underlying "principal components" that explain most of the variance in hand shape and size. This allows designers to focus on a few key measurements instead of dozens. Studies have used this approach to determine that a small number of parameters, such as grip circumference and palm breadth, can be used to derive many other dimensions of a prosthetic hand.
- **Regression Models:** Linear regression is used to establish mathematical relationships between different body measurements. For example, models can be created to predict finger lengths based on hand length or stature, which is useful when only partial measurements are available. One study found a significant positive correlation between the length and breadth of the proximal phalanx and a person's stature, though another study found a weak correlation between index finger length and width.
- **Allometric Scaling:** This involves studying the proportional relationships between the sizes of different body parts. These scaling laws can be used to estimate unmeasured dimensions from known ones, helping to create a proportionally correct model even from limited data. This is particularly relevant when attempting to reconstruct a missing hand based on the dimensions of the sound limb or other body parts.

Identified Gaps and Recommendations

Despite advancements in measurement technology and design workflows, the literature reveals several persistent gaps in the field of upper limb anthropometry for prosthetics.

- **Lack of Amputee-Specific Data:** Most available anthropometric databases are based on able-bodied populations. The morphology of a residual limb can differ significantly from a non-amputated limb due to surgical procedures, muscle atrophy, and tissue changes. There is a critical need for more data collected directly from amputees to inform better socket design and overall prosthetic fitting.
- **Data Scarcity for Pediatric Populations:** Children with limb differences require new prostheses frequently as they grow, yet there is a significant lack of comprehensive anthropometric data for this group. Developing robust growth models and collecting longitudinal data on children would greatly improve the design of pediatric

prostheses.

- **Standardization of Measurement Protocols:** While standards like ISO 7250 exist, there is a wide variety of measurement techniques and landmarks used across different studies. This inconsistency makes it difficult to compare data across populations and establish universally applicable design principles. A more standardized and widely adopted measurement protocol is needed.

Based on these gaps, the following recommendations are proposed:

1. **Establish Comprehensive Amputee Databases:** Initiate large-scale data collection efforts focused on the anthropometry of residual upper limbs, categorizing data by amputation level, age, and sex.
2. **Conduct Longitudinal Pediatric Studies:** Fund and conduct studies that track the anthropometric development of children with and without limb differences to create predictive growth models for prosthetic design.
3. **Develop a Minimum Essential Measurement Set:** Researchers and clinicians should collaborate to define a standardized minimum set of key anthropometric parameters that are essential for the design and fitting of different types of upper limb prostheses. This would improve data comparability and streamline the design process.
4. **Promote Open-Access Data:** Encourage the sharing of anonymized anthropometric datasets to accelerate research and innovation in the field, allowing designers and engineers worldwide to develop more effective and accessible prosthetic solutions.

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Measurement Taxonomy by Body Region

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HAND

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- * **Hand Length:** Defined as the distance from the wrist crease (stylium landmark) to the tip of the middle finger.
- * **Hand Breadth:** Measured across the metacarpal joints (metacarpal II to metacarpal V), this dimension represents the widest part of the hand.
- * **Palm Length:** This is the distance from the base of the middle finger to the wrist crease and is essential for determining the length of the prosthetic palm.
- * **Hand Circumference/Girth:** Measured around the metacarpals, this parameter is important for socket fit and prosthetic volume.
- * **Hand Thickness:** The measurement through the hand at the metacarpals is used to ensure the prosthetic can accommodate the user's anatomy.

FINGERS

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- * **Individual Finger Lengths:** The lengths of the index, middle, ring, and little fingers are measured to determine the reach and dexterity of the prosthesis.
- * **Phalanx Lengths (Proximal, Middle, Distal):** The lengths of the individual bone segments of each finger are critical for replicating the natural range of motion.

* **Finger Breadth and Depth:** Measured at the proximal and distal interphalangeal joints, these dimensions are critical for ensuring a proper fit.
* **Interphalangeal Joint Spacing:** The distances between the DIP, PIP, and MCP joints are derived from these measurements.

THUMB

The thumb's unique geometry and opposition are vital for dexterity, making its accurate measurement a priority.

* **Thumb Length:** Measured from the thumb tip to its base, this is a key parameter for ensuring effective grip.
* **Thumb Phalanx and Metacarpal Lengths:** The lengths of the thumb's bone segments are used to model the joint.
* **Thumb Breadth:** This measurement informs the overall size and shape of the thumb mechanism.
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* **Forearm Circumference:** Measurements at multiple points along the forearm (proximal, mid, distal) are used to create a well-fitted socket that distributes pressure evenly and provides stable suspension.
* **Cross-sectional Shape:** Modern workflows use 3D scanning or CT/MRI to capture the non-circular cross-sectional shape of the residual limb, which is critical for creating a comfortable and functional fit.

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CT/MRI	Medical imaging techniques that provide detailed internal and external anatomical data.	Highly accurate, captures internal structures.	Expensive, requires specialized equipment and expertise.
Radiography	X-ray imaging used to measure bone lengths and joint geometry from digitized images.	Accurate for bone measurements.	2D projection, limited to bone structure.

| **Photogrammetry** | Uses multiple photographs from different angles to reconstruct a 3D model or take
processing. |
| **Image Processing** | Software-based methods using algorithms (e.g., OpenCV, Mediapipe) to detect 1

Population Variability and Databases

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- Parametric CAD:** This is a primary application where key anthropometric values (e.g., hand length, wrist circumference) are used to define the geometry of a prosthetic limb.
- Socket Design:** For transradial and transhumeral prostheses, the design of a comfortable and stable socket is critical, often relying on measurements of residual limb volume and shape.
- Linkage Mechanism Sizing:** The lengths of phalanges and the distances between joints directly influence the design of prosthetic finger or wrist mechanisms, used to replicate finger movement. Accurate measurements ensure that the prosthetic finger can reach and grasp objects effectively.
- 3D Printing Customization:** Additive manufacturing enables the direct fabrication of these parametrically defined components, allowing for highly customized prostheses.
- Finite Element Analysis (FEA):** Anthropometric data is used to create anatomically correct models of the limb interface, allowing designers to optimize the structural integrity and comfort of the prosthesis by simulating stresses and strains.
- Kinematic Modeling:** Measurements of bone lengths and joint locations extracted from sources like motion capture or imaging are used to model the range of motion and functional requirements of the prosthetic limb.

Statistical Approaches

To manage the complexity of the numerous variables involved in hand anthropometry, researchers employ statistical methods to analyze and synthesize the data.

- Factor Analysis and Principal Component Analysis (PCA):** These methods are used for dimensionality reduction, identifying the most significant variables that explain the variance in the data.
- Regression Models:** Linear regression is used to establish mathematical relationships between different anthropometric measurements, allowing for the prediction of one measurement based on others.
- Allometric Scaling:** This involves studying the proportional relationships between the sizes of different body parts, which is crucial for designing prostheses that scale appropriately across different population groups.

Identified Gaps and Recommendations

Despite advancements in measurement technology and design workflows, the literature reveals several persistent gaps and challenges in the field of upper limb prosthetics.

- Lack of Amputee-Specific Data:** Most available anthropometric databases are based on able-bodied populations. The morphology of a residual limb can differ significantly from a non-amputated limb due to surgical procedures, muscle atrophy, and tissue changes. There is a critical need for more inclusive data collection that specifically addresses the needs of amputees.

- * **Data Scarcity for Pediatric Populations:** Children with limb differences require new prostheses
- * **Standardization of Measurement Protocols:** While standards like ISO 7250 exist, there is a wide v

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